Binance Gold & Silver Perpetuals and ETF Markets

Event Study on Volatility, Liquidity, and Integration

Course: Data Analysis and Visualization in Python

Group 4:
Yaroslav Kuzmin, Yevhen Nezghovorov, Hlieb Nikoniuk, Tymur Tleuberlin

References:

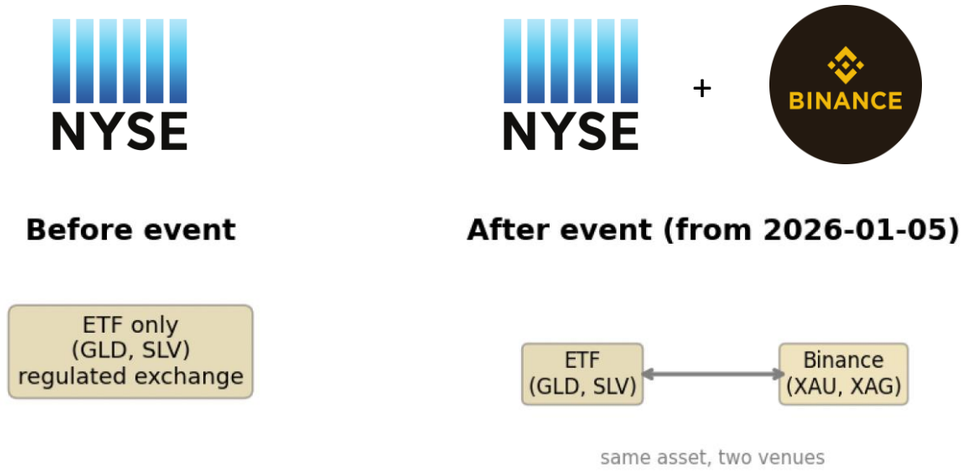
•Corbet, Shaen, Brian Lucey, Maurice Peat, and Samuel Vigne. "Bitcoin Futures—What Use Are They?." *Economics Letters* 172 (2018): 23-27
•Aminhud, Yakov. "Illiquidity and Stock Returns: Cross-Section and Time-Series Effects." *Journal of Financial Markets* 5, no. 1 (2002): 31-56

Research Question & Why That Matters

Gold and silver are traded as **ETFs** (GLD, SLV) on regulated exchanges. In early 2026,Binance started listing **perpetuals** on the same assets (XAUUSDT, XAGUSDT). So now there are **two places** to get exposure to the same thing.

We ask: did the **market change** after that? More volatile? More trading? More linked to the new venue?

This matters for traders, market-makers, and regulators.



> *"The introduction of derivatives can change spot markets—volatility, liquidity, and how prices move together."*
> Augustin, Rubtsov & Shin (2023), Management Science — Bitcoin futures and spot markets.

Literature overview

What was studied	Methods	Results (key)
Augustin, Rubtsov & Shin (2023), Management Science: introduction of **Bitcoin futures** → impact on **spot/cash** markets (exchanges with vs without futures).	**DiD** (treatment = BTC-USD pairs with futures; control = other pairs). Fixed effects (exchange pair, month, currency). Event-time dynamics. **Outcomes:** volatility, volume, Amihud, Roll, price synchronicity/integration. Winsorization; clustered SE.	Derivatives change spot: integration, liquidity, volatility.
"The influence of ETFs on the price discovery of gold, silver and oil", Journal of Economics and Finance (2013): how **ETFs interact with spot and futures** in commodity markets.	**Price discovery metrics** (e.g. information share / VECM-based methods) to see which market (spot, futures, ETF) leads; time-series models on returns and spreads.	ETFs are part of price discovery: for gold/silver they share or sometimes lead price adjustment with futures/spot. Shows that adding a new trading venue can shift where prices are set.

We orient on the "reverse" setting: literature on how **derivatives affect spot markets**—e.g. Bitcoin futures vs spot. Our case: new perpetuals (Binance) vs incumbent ETFs (GLD, SLV).

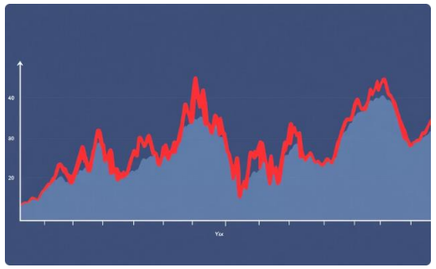
What we take for our study: Event-study framing; same type of outcomes (volatility, volume/liquidity, integration); controls (DXY, VIX, 10Y) to isolate the event; winsorization; HAC (Newey–West) SE; pre/post comparison and OLS with post_event.

What we cannot take: DiD—we have no control group (only GLD/SLV before/after). **Pre-event integration**—Binance did not exist before the launch, so we cannot test "integration increased"; we only describe post-event correlation/level.

Hypothesis

Main Research Question: Did the launch of Binance gold and silver perpetual futures (January 5, 2026) change the behavior of traditional ETF markets (GLD, SLV)?

Event date: January 5, 2026



H1: Volatility

ETF volatility increased after the Binance perpetual listing.

H2: Trading Volume

ETF trading volume increased after the listing.

H3: Market Integration

ETF returns became more integrated with Binance perpetual returns post-launch.

Economic Question & Methodology

Data Sources: Binance API (1-minute, 67k+ observations), Yahoo Finance (daily ETFs), FRED (10Y Treasury) - 87 variables

Source	What we get	Frequency	Rows	Note
Binance API	XAUUSDT, XAGUSDT (OHLCV), funding, OI	1m	67k+	From 2026-01-05 (XAU), 2026-01-07 (XAG)
Yahoo Finance	ETFs: GLD, SLV. Controls: DXY, VIX, TNX. GLD/SLV 1m	Daily; 1m last 7d	118 daily	1m only last ~7 days
FRED	DGS10 — 10Y Treasury (macro control)	Daily	in daily panel	Same role as TNX

Methods:

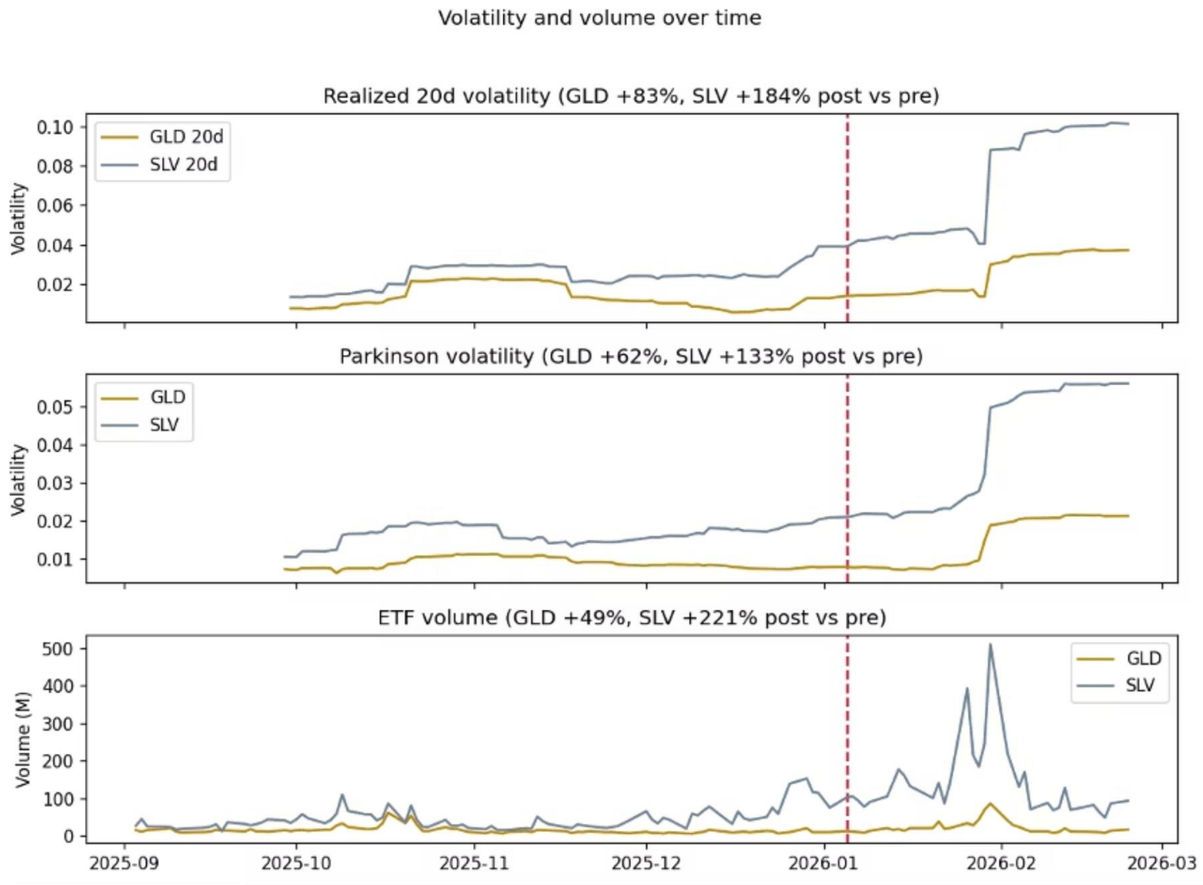
- OLS regressions that controlled for macroeconomic factors like the VIX, DXY, and Treasury yields to isolate the true impact of the Binance
- Amihud Illiquidity Ratio to measure actual trading stability rather than just raw volume
- Price correlations to determine how deeply these traditional and crypto markets integrated
- We used a T-test to make sure our findings were real and statistically significant

Regression & T-Test Results

Metric	Pre Mean	Post Mean	% Change	t-statistic	t-test p-value	PostEvent β	Regression p-value
GLD 20d vol	0.013	0.025	+83%	5.809	<0.001	0.011	0.008
SLV 20d vol	0.024	0.069	+184%	9.492	<0.001	0.044	<0.001
GLD volume	15.124	22.480	+49%	2.377	0.022	—	—
SLV volume	43.589	140.791	+221%	5.913	<0.001	—	—
Corr GLD–XAU		0.918					
Corr SLV–XAG		0.935					

Volatility increases remain statistically significant after controlling for DXY, VIX, and 10Y yield.

Dynamic Volatility & Volume Evidence



Structural Break After Launch

• Clear upward shift in realized volatility after January 5, 2026

• Silver response substantially stronger than gold

Magnitude of Change

• GLD 20d volatility: +83% (post vs pre)

• SLV 20d volatility: +184% (post vs pre)

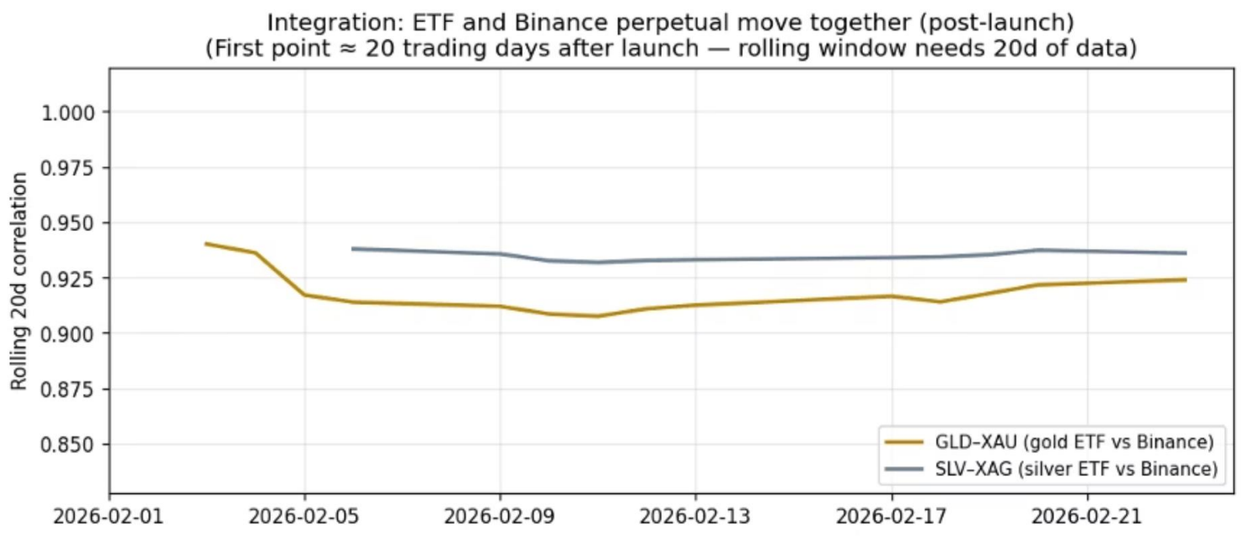
• ETF trading volume increases, especially for silver (+221%)

Timing Consistency

• Shift occurs immediately after Binance perpetual launch

• No similar discontinuity observed pre-event

Rolling Correlation Dynamics



Persistent Post-Launch Integration

• Rolling 20-day correlations stabilize around 0.92–0.94

• Gold and silver ETFs move closely with Binance perpetuals

Evidence of Unified Price Discovery

• High co-movement suggests integrated markets

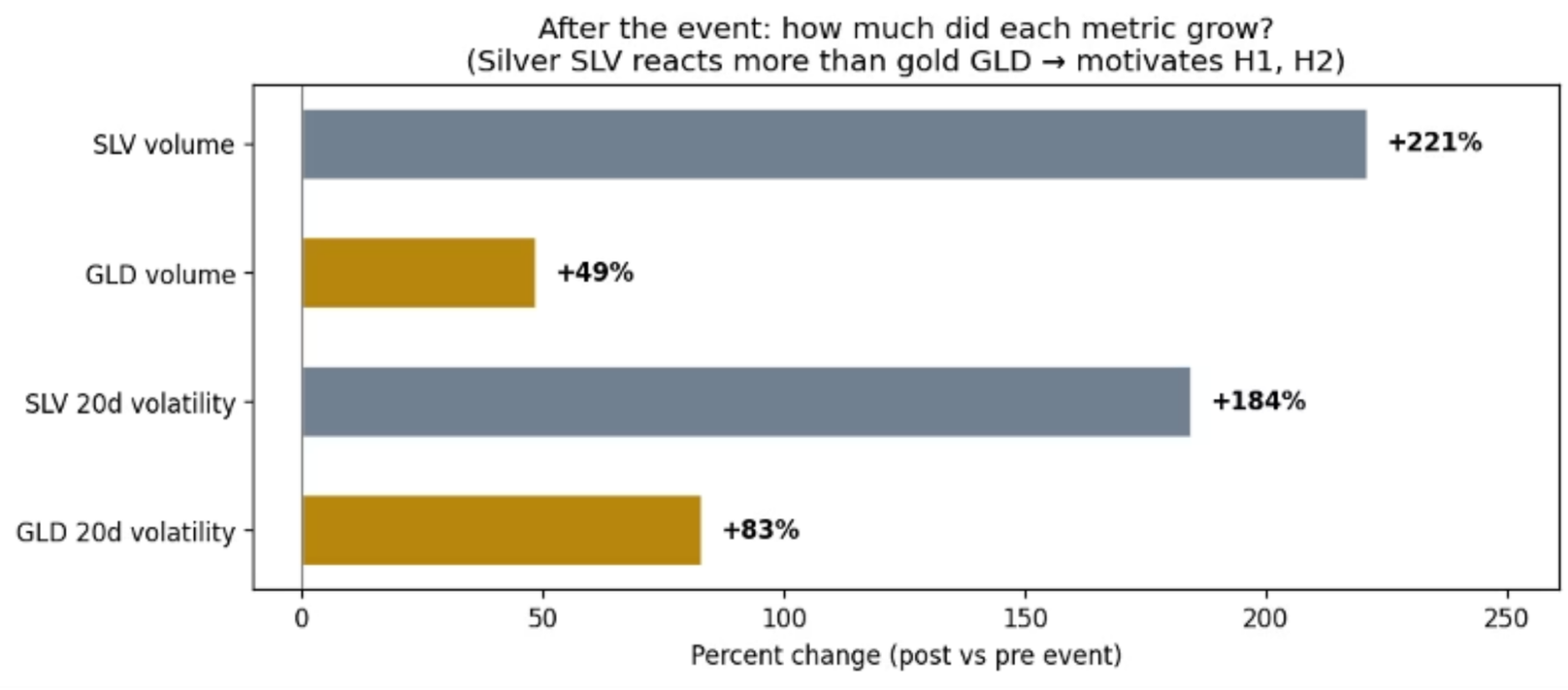
• Arbitrage likely links ETF and perpetual pricing

Dynamic Stability

• Correlation remains elevated throughout post-event window

• No evidence of decoupling after initial adjustment

Magnitude of Post-Event Effects



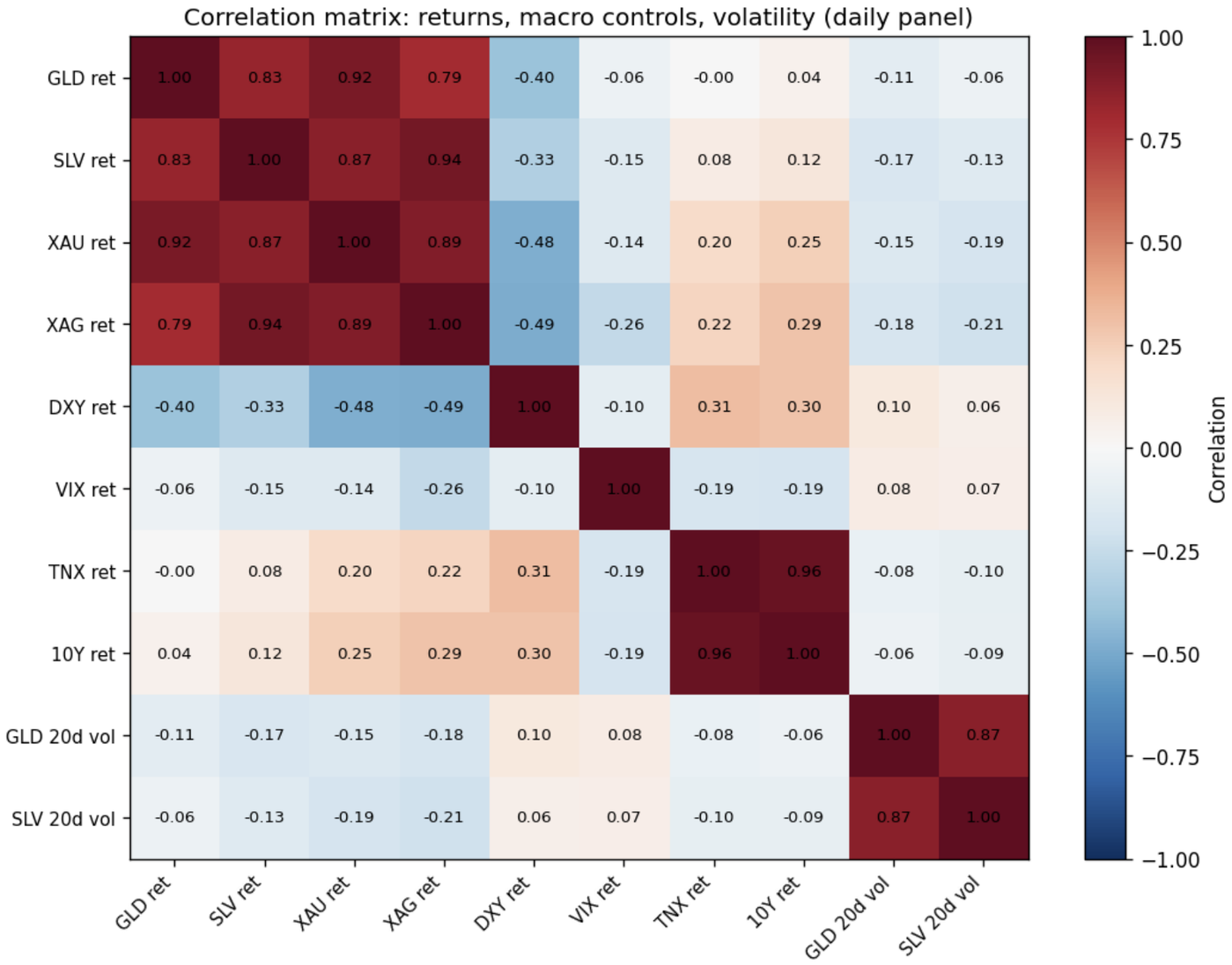
Asymmetric Impact

- Silver exhibits substantially larger response than gold
- Volume effects exceed volatility effects

Economic Pattern

- Liquidity expansion strongest in SLV (+221%)
- Consistent with increased cross-venue arbitrage activity

Correlation matrix



Pairwise correlations between the main variables we use: ETF returns(GLD, SLV), Binance perpetual returns (XAU, XAG), macro controls (DXY, VIX,10Y/TNX), and realized volatility (GLD 20d, SLV 20d).

High correlation between GLD and XAU (and SLV and XAG) in the post-event overlap supports integration; macro variables help us control for common factors inregressions.

Conclusions & Interpretations

GLD +83%, SLV +184%

Volatility ↑

GLD +49%, SLV +221%

Volume ↑

Post-launch correlation ≈ 0.92–0.94

Integration High

Higher Volatility ≠ Disorder

May reflect increased hedging, arbitrage activity, cross-venue informed trading

Limitations & Future Research

No Pre-Event Binance Data

Cannot test correlation increase

No Control Group

No difference-in-differences design

Short Post Window

Longer sample needed